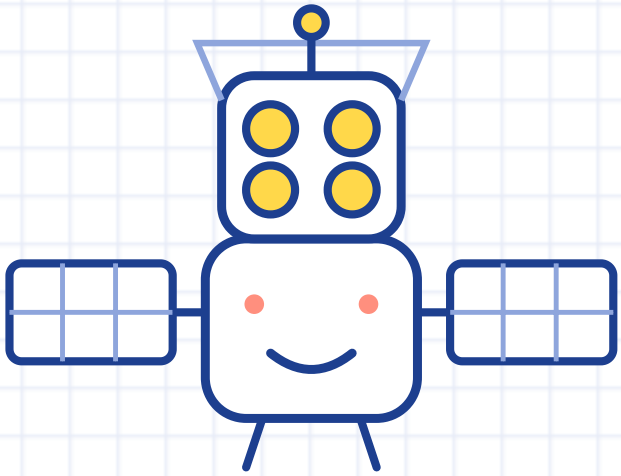
Field file, October 2026

Tess and the two impostors

An autonomous bot searched more than 5,000 stars in NASA TESS data for undiscovered planets. Its two best finds turned out to be pairs of stars in disguise. This is how they fooled it, and how it caught them.



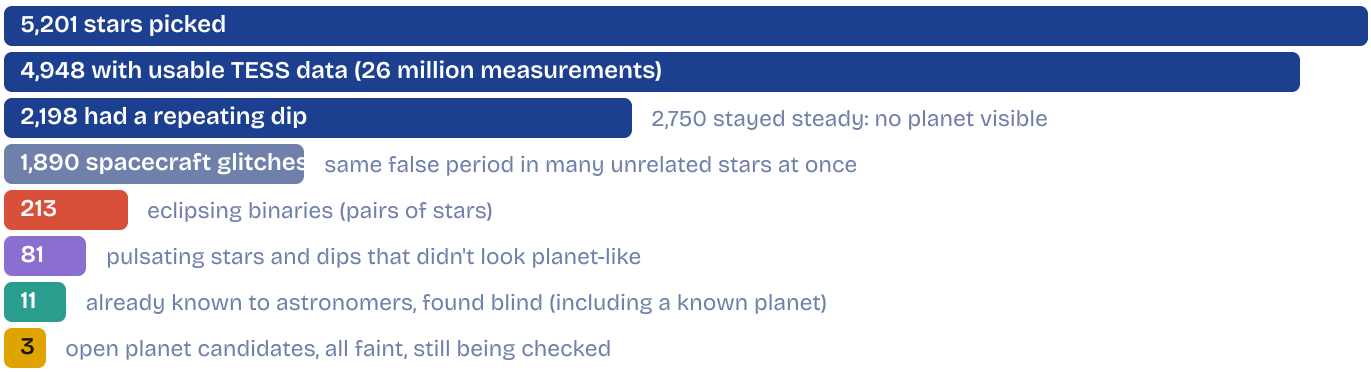
Alejandro Valdez
Aerospace Engineering, University of Central Florida

alexvaldex.github.io/nasa-tess-planet-search
github.com/alexvaldex/nasa-tess-planet-search

Logbook

What 5,201 stars turned up

Tess wakes up on GitHub's free computers every few hours, picks stars NASA's TESS satellite photographed, downloads their brightness measurements, and searches each one for a dip that repeats on a schedule. Every repeating dip then has to survive a gauntlet of tests built to catch impostors. Here is where every star ended up after Tess's first two days.



The twist

Tess's two strongest early candidates, TIC 267889592 and TIC 415954197, passed every test she had. Then a deep dive through every sector TESS ever took of them, up to six years of data, showed both are eclipsing binaries. Each one exposed a blind spot in Tess's vetting. Both blind spots are now fixed, and four more look-alikes were caught the same way.

Can Tess find a real planet?

Tested blind on 281 stars with confirmed TESS planets, she found **241 (86%)**. On the hardest fakes there are, signals that passed NASA's own first cut and were later disproven, she rejects **70%** while keeping 78% of real planets.

Open case: TIC 374098817

A dip of 0.03% every **19.9 hours**, still there across 2019, 2022 and 2024 data (signal-to-noise 13). If it is a planet it would be about **2.8 times Earth's size**, orbiting an F-type star in Andromeda about **1,100 light-years** away, so close that its surface would be around **2,500 K**. Planets that size so close to their star are rare, which makes it interesting and also a reason for extra suspicion.

Running cost

$0. Up to 800 stars per run, every few hours, on GitHub Actions. All code and results are public.

Case file 1

ECLIPSING BINARY

TIC 267889592

Constellation Carina · RA 07h 41m 01s, Dec −51° 10′ 32″ · about 926 light-years away

Why it looked like a planet

Every 19.054 days the star dimmed by about 1.5% for roughly four hours, then went back to normal. Tess saw it happen four times in 2019, at the same depth each time. For a star three-quarters the size of the Sun, a 1.5% dip is exactly what a Jupiter-sized planet would cause, and a four-hour crossing is exactly how long a planet on a 19-day orbit should take. There was no dip halfway round the orbit, the dips were the same depth whether odd or even, and the star's image didn't shift on the detector. Tess's AI model scored it 0.80 out of 1.

What gave it away

The deep dive pulled every TESS sector of this star: 2019, 2021, 2023 and 2025. Twelve clean dips (plus one partial) lined up perfectly on a **19.0542-day** period, which was good news. But folding six years of data over one orbit revealed a **second dip of almost the same depth at phase 0.62**, about 11.7 days after the first. A planet blocks starlight once per orbit. Two equal dips per orbit means two similar stars taking turns passing in front of each other.

The second dip isn't halfway round the orbit, which tells you the orbit is oval. From the timing alone, $e\cos\omega \approx (\pi/2)(0.615 - 0.5) \approx$ **0.18**. The dips are shallow and V-shaped because the two stars only graze each other's edges as seen from Earth.

Why nobody had flagged it

- It was never on NASA's two-minute priority list. TESS only caught it in its wide-field images, every 30, 10 and later 3.3 minutes.
- With a 19-day orbit, each 27-day TESS sector shows at most a couple of each kind of eclipse, which is too few for one-sector-at-a-time searches.
- The catalog of TESS eclipsing binaries (Prša et al. 2022) covered only priority targets from TESS's first two years. Gaia DR3 doesn't list it as a variable star, and it isn't in the Variable Star Index (VSX) or SIMBAD.
- The eclipses are only about 2% deep, last around seven hours, and come every 19 days. Ground surveys that check a star every few nights easily step over them.

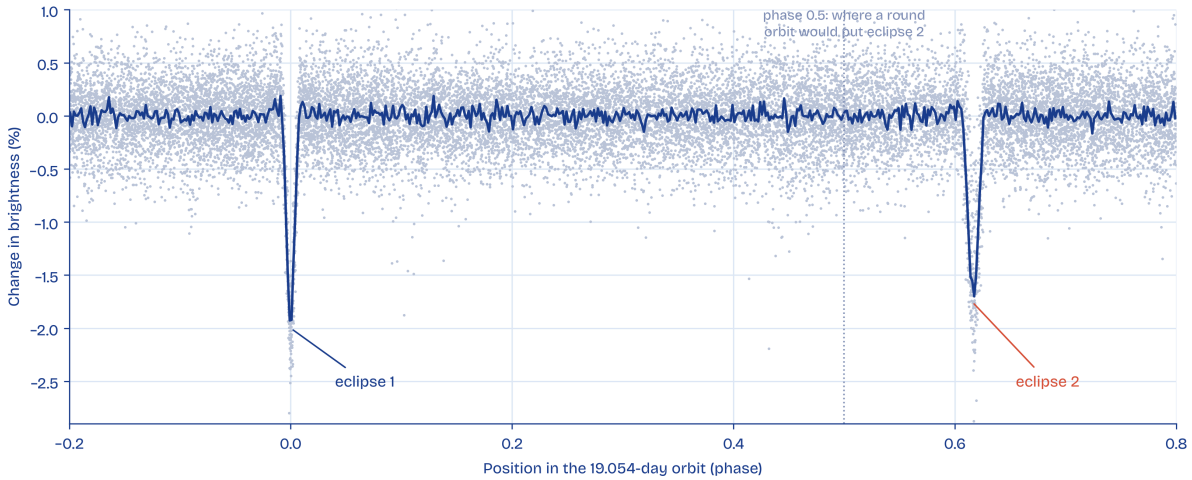
Verdict	eclipsing binary
Orbit	19.0542 days (±4 seconds)
Eclipse depths	~1.8% and ~1.6%
Second eclipse	phase 0.615 (oval orbit)
Eclipses recorded	12 clean primaries plus their partners, 2019 to 2025
Distance	284 pc ≈ 926 light-years (Gaia parallax 3.522 mas)
Brightness	V = 13.6 (TESS mag 12.8)
Colour / temperature	orange, ~5,000 K
Speed away from us	58.6 km/s (Gaia)
Catalogs checked	8,151 TOIs, 5,138 CTOIs, 6,138 planets, TESS & Gaia EB catalogs, VSX, SIMBAD: not listed

A crowded pixel

Another, unrelated eclipsing binary (Gaia DR3 5493043445356438400, period 0.31 days) sits just **5.7 arcseconds** away. Each TESS pixel is 21 arcseconds wide, so both binaries share a pixel. Its period doesn't match, so it isn't the source of these eclipses.

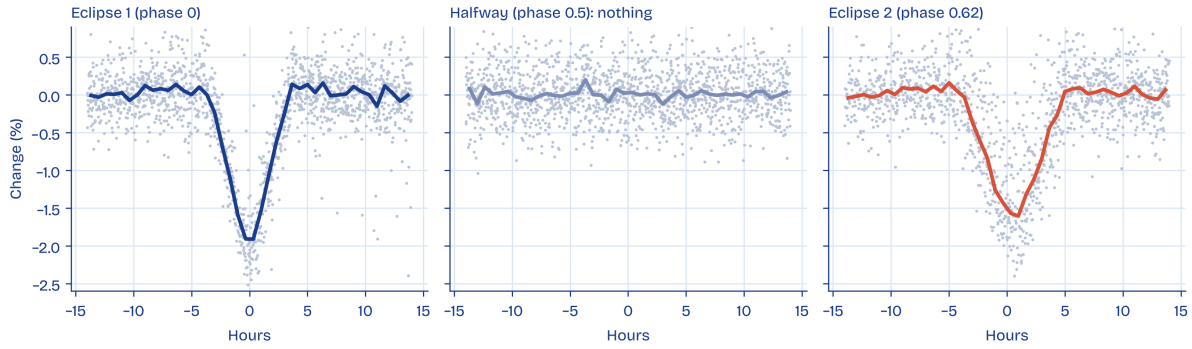
Case file 1, evidence

Six years of eclipses, folded over one orbit



Every TESS measurement from 2019 to 2025, placed by where it falls in the 19.054-day orbit. A planet would make one dip. There are two.

The two eclipses up close



The dotted middle panel is where a round orbit would put the second eclipse. The real second eclipse arrives later, at phase 0.62.

Fun facts

Clockwork. Tess timed 12 eclipses across 117 orbits. The period is 19.05420 days, known to about 4 seconds.

Old light. The light TESS recorded left these stars around the year 1100.

Hard to see from Florida. At −51° declination it never climbs more than about 10° above Orlando's southern horizon. Best on February evenings.

Grazing. Two similar stars eclipsing head-on would dim by up to half. These dim by about 2%, so they only clip each other's edges.

Next eclipse. Predicted for Sunday, November 1, 2026 around 2:58 a.m. Eastern, give or take about 15 minutes.

Very close pair. If both stars are about 80% of the Sun's mass, they're only about 0.16 AU apart: 2.4 times closer than Mercury is to the Sun.

Faint. About 1,000 times too faint for the naked eye. You'd need roughly an 8-inch telescope under dark skies.

A long trip. At Voyager 1's speed (17 km/s) the journey would take about 16 million years.

What would settle it for good

Two or three spectra would show the lines of two stars moving back and forth every 19 days. A ground-based telescope timed for November 1 would confirm the eclipse belongs to this star, not its neighbour.

Period from a linear fit to 12 eclipse times; next-eclipse uncertainty from the fit

Case file 2

ECLIPSING BINARY

TIC 415954197

Constellation Pegasus · RA 22h 49m 27s, Dec +11° 05′ 48″ · about 1,800 light-years away

Why it looked like a planet

In 2024 TESS data, Tess found a 1.2% dip every 7.140 days, three times in one month, each lasting about four hours. Around a star 1.3 times the Sun's size, that matches a planet a bit larger than Jupiter, a kind of world called a hot Jupiter. The dips had a flat-ish bottom and nothing showed up halfway round the orbit. Her AI model gave it 0.99 out of 1, her highest score so far.

What gave it away

Two things happened at once. First, TESS had also watched this star in **2022**, and Tess hadn't used that data. Second, re-measuring by hand showed that exactly **half an orbit** after each 1.2% dip there's a much deeper **3.2%** dip. It appears in both years, seven times in total, at signal-to-noise 113.

A planet is dark, so it can't make a second dip, let alone one deeper than the first. Two different eclipses half an orbit apart means two stars: the deep eclipse is when the cooler companion passes in front of the main star, and the shallow one is when the companion goes behind it. Tess's halfway check had been reading a value from a software library that came back with an error of zero, which made a 54-sigma signal look like nothing. She now measures it herself.

Why nobody had flagged it

- TESS has only looked at this patch of sky twice, in 2022 and 2024, and only in its wide-field images. The star wasn't on the priority list.
- It came after the observing years covered by the TESS eclipsing-binary catalog. Gaia DR3 doesn't list it as variable, and it isn't in VSX or SIMBAD.
- Candidates from recent TESS sectors take time to work through, and faint stars wait longest.

Verdict	eclipsing binary
Orbit	7.1399 days
Deep eclipse	3.2% (signal-to-noise 113)
Shallow eclipse	1.2% at phase 0.50
Orbit shape	round (second eclipse exactly halfway)
Data	TESS sectors 56 (2022) and 83 (2024)
Distance	555 pc ≈ 1,810 light-years (Gaia parallax 1.802 mas)
Brightness	V = 13.5 (TESS mag 12.8)
Main star	yellow-orange, ~5,400 K, ~1.3× the Sun's size
Catalogs checked	8,151 TOIs, 5,138 CTOIs, 6,138 planets, TESS & Gaia EB catalogs, VSX, SIMBAD: not listed

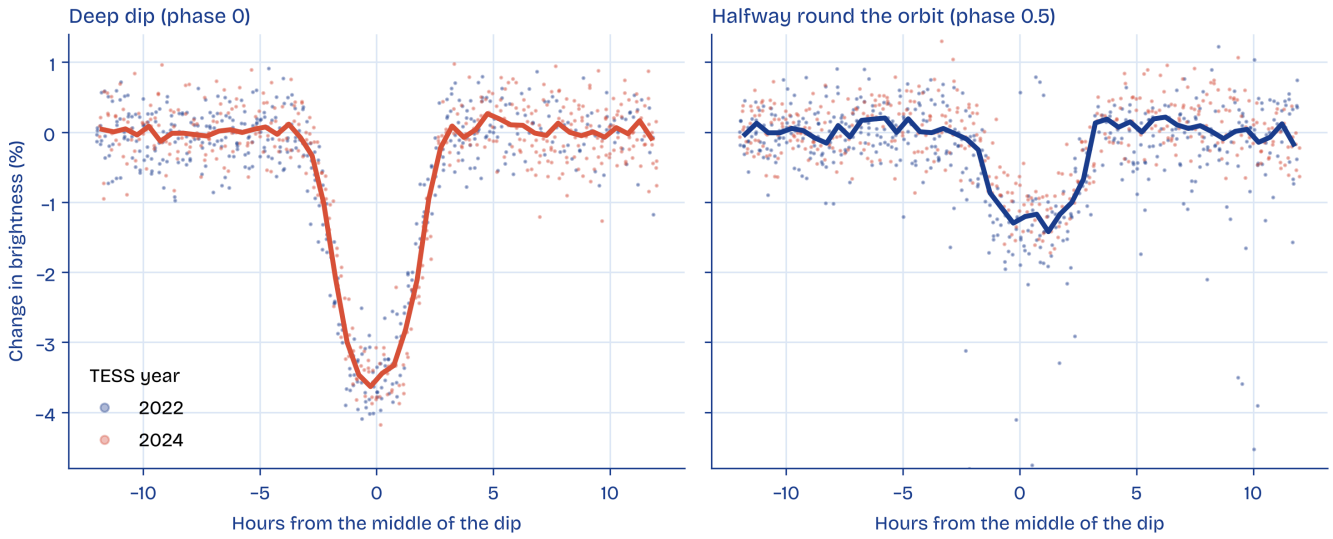
Whose eclipses?

A brighter, unrelated star sits 56 arcseconds away, about 2.6 TESS pixels. The TESS data for this star has no information on where the light is coming from, so a ground-based telescope is needed to confirm which of the two stars is eclipsing.

Data: NASA TESS (QLP), Gaia DR3, TIC v8.2

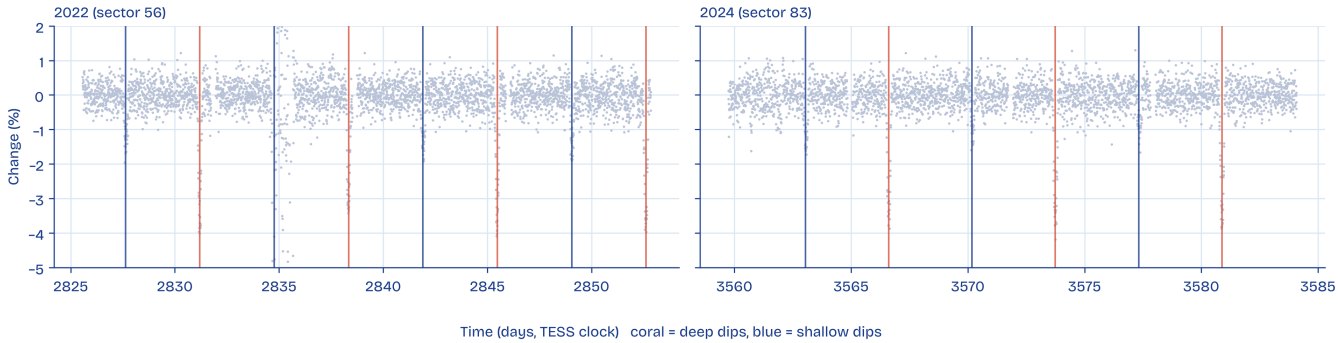
Case file 2, evidence

The giveaway: a second, deeper eclipse half an orbit later



Both TESS years folded on the 7.1399-day orbit. Tess first locked on to the shallow eclipse (right). The deep one (left) sat exactly half an orbit away.

Both TESS visits, eclipses marked



Coral marks the deep eclipses, blue the shallow ones. Same pattern in 2022 and 2024.

Fun facts

Up tonight. Pegasus is high in Orlando's south-eastern sky on October evenings, near the Great Square. The star reaches about 73° up.

Tidy orbit. Pairs of stars closer than about ten days usually have their orbits made round by tides. This one has. Case file 1, at 19 days, hasn't.

Cooler partner. The shallow eclipse is 38% as deep as the deep one, so the companion's surface is much dimmer in TESS's red light: a cooler, smaller star.

Roman-era light. The light TESS caught left this system around 200 AD.

Snug. The two stars are only about 0.08 AU apart, almost five times closer than Mercury is to the Sun.

A long trip. At Voyager 1's speed the journey would take about 32 million years.

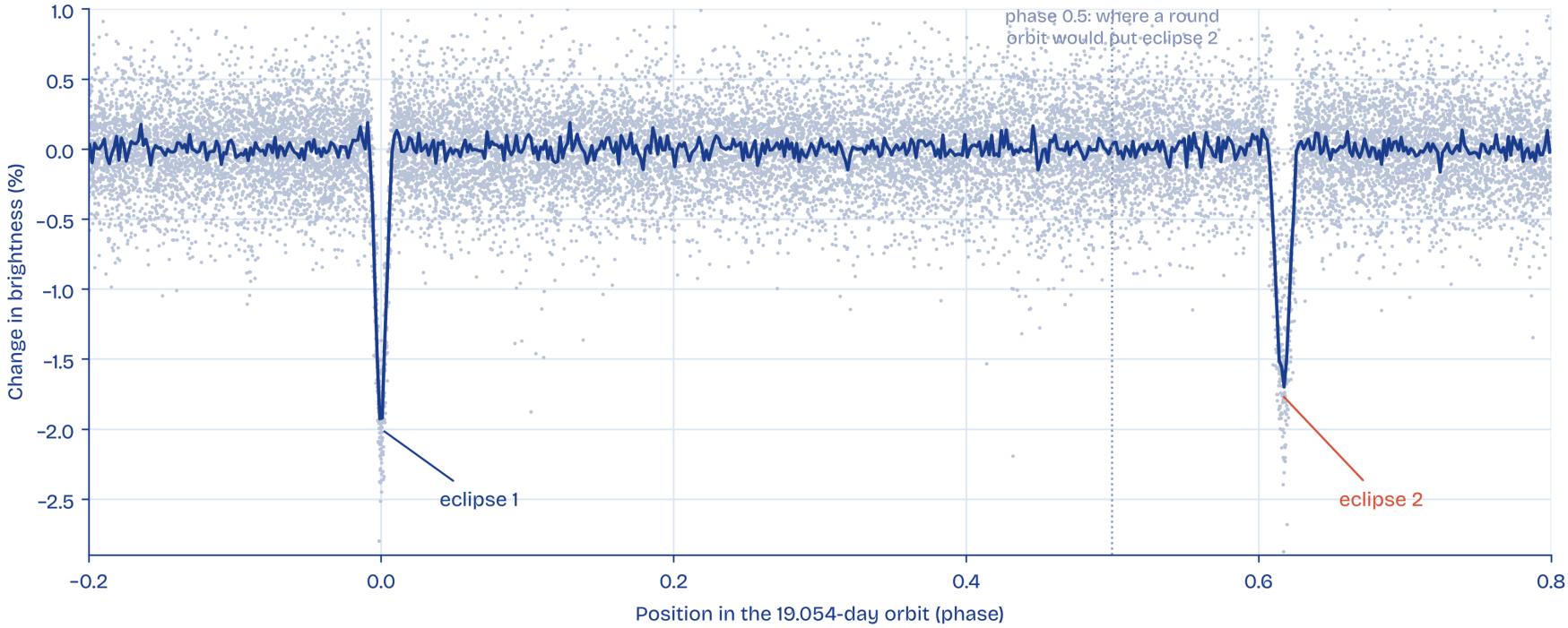
The lesson

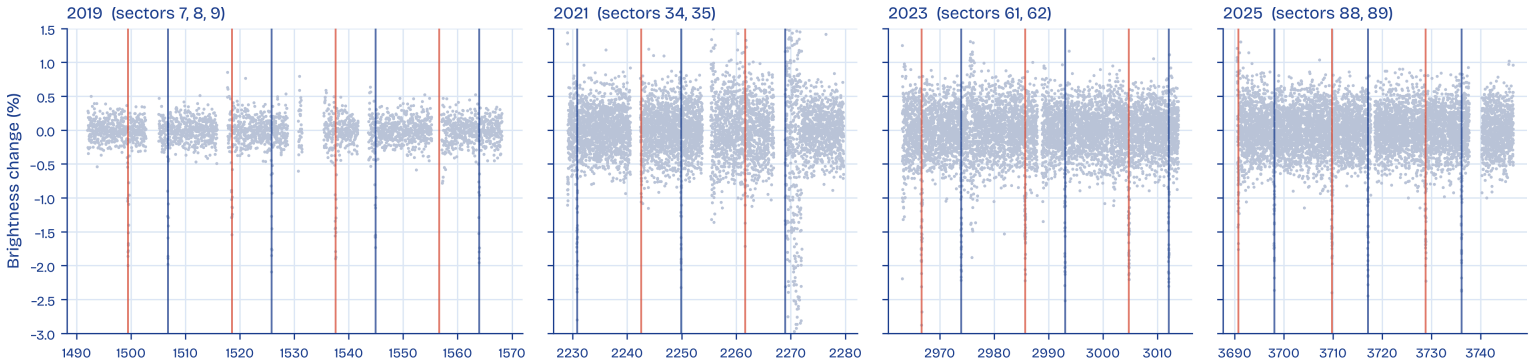
The AI model was confident, and wrong. The fix wasn't a smarter model. It was an older dataset and a test that measured the physics directly. That's why Tess now pulls every TESS sector of any star before calling it a candidate.

Separation assumes masses of about 0.95 and 0.6 solar masses

TIC 267889592

Two eclipses per orbit, six years running

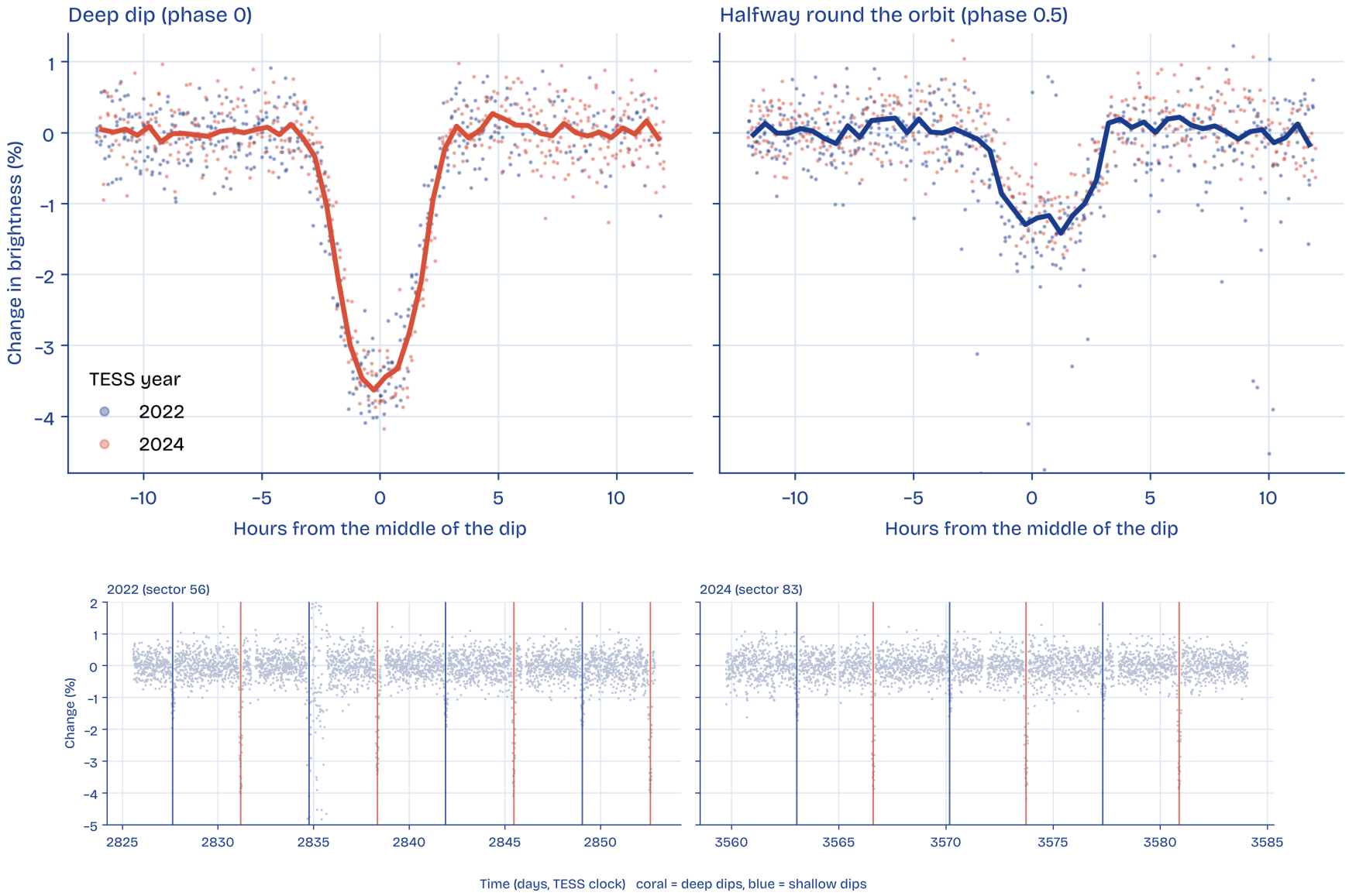




Every TESS measurement of TIC 267889592, 2019 to 2025. Top: folded on the 19.0542-day orbit. Bottom: each TESS visit, blue = eclipse 1, coral = eclipse 2.

TIC 415954197

A planet can't make two dips



TESS 2022 and 2024. The 1.2% dip Tess first flagged (right) sits exactly half an orbit after a 3.2% eclipse (left): two stars, not a planet.

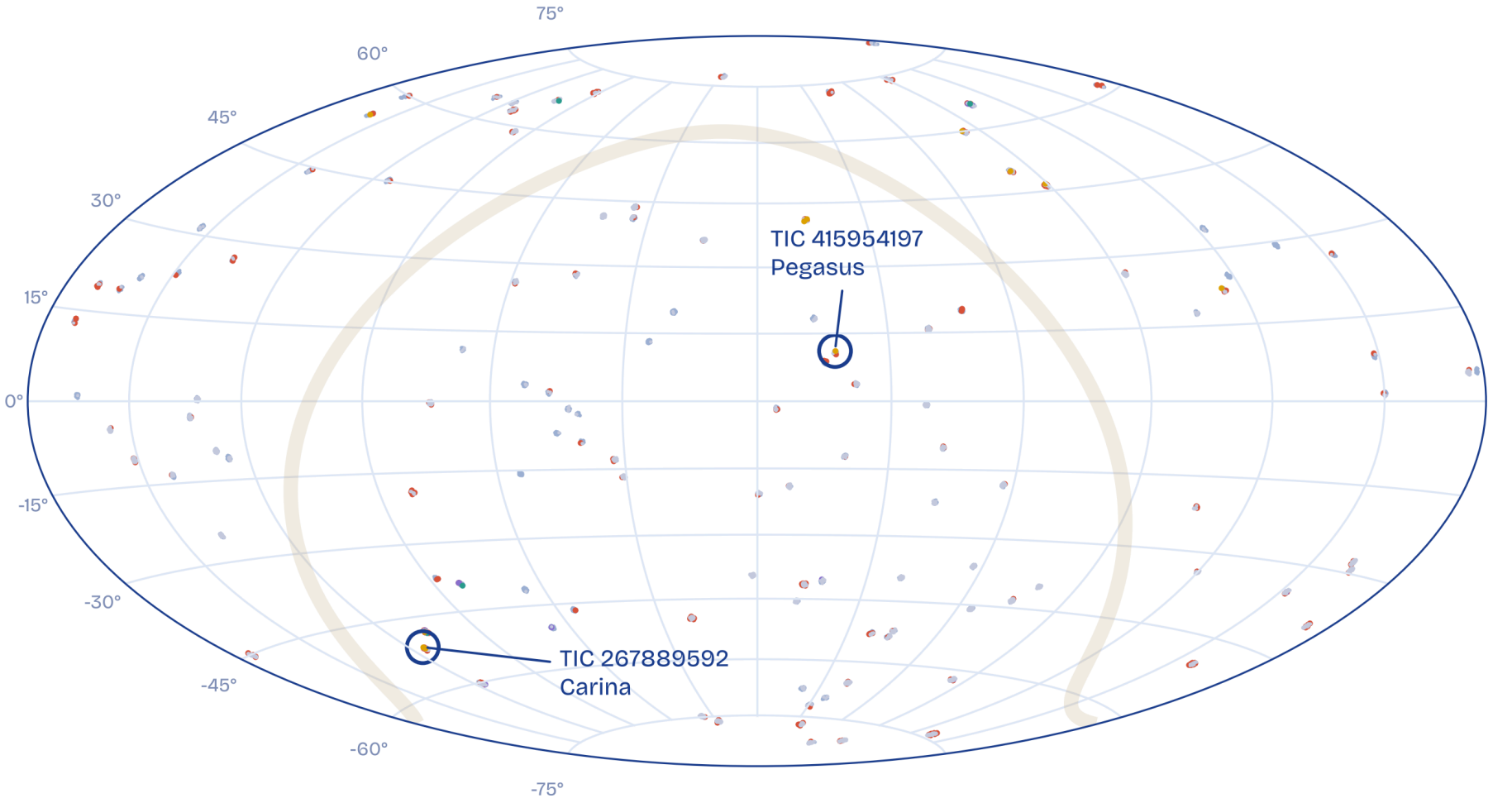
Tess field file

Coverage

Where Tess has looked

Every star Tess has checked: 4,948

Hammer projection of the whole sky. The tan band is the Milky Way, which Tess skips because stars there blend together.



Each dot is one star. Circled: the two case-file binaries, in Carina and Pegasus.

Field guide

Planet, or two stars?

A planet crossing its star and a star grazing its partner can make dips that look the same. The difference is in what happens during the rest of the orbit.

Planet



Pair of stars, round orbit



Pair of stars, oval orbit



The physics behind each test

$\delta \approx (R_p / R_\star)^2$

Dip depth tells you the size ratio. A 1% dip on a Sun-sized star is a Jupiter.

$a^3 = M_\star P^2$

Kepler's third law (a in AU, M in solar masses, P in years): orbit size from period.

$T \approx (P/\pi) \cdot \arcsin(R_\star / a)$

How long a central transit should last. Too short or too long is suspicious.

$P_{transit} \approx R_\star / a$

The chance an orbit happens to be lined up for us to see transits.

$e \cos \omega \approx (\pi / 2)(\varphi_2 - 0.5)$

If the second eclipse isn't at phase 0.5, the orbit is oval.

$T_{eq} = T_\star \sqrt{(R_\star / 2a)}\,(1 - A)^{1/4}$

A planet's temperature from its star and distance (A is how reflective it is).

$d\,[pc] = 1 / p\,[arcsec]$

Distance from Gaia's parallax measurement.

$SNR \approx (\delta / \sigma)\sqrt{N_{in}}$

Signal-to-noise: deeper dips and more points during the dip help.

Lab log

Every mistake Tess made, and the fix

Real data is humbling. Each problem below turned up in Tess's own results and became a new test.

- **Oct 12, first real run.** More than half the "signals" were TESS itself: the same false periods (5.9, 8.4, 13 days) showed up in unrelated stars. *Fix: throw out any period shared by several stars observed at the same time.*
- **Oct 12.** One star "dipped" 37% because of a handful of bad pixels. *Fix: a robust depth check, and remove lone bad points.*
- **Oct 12.** A single slow NASA download froze a whole run. *Fix: 90-second timeouts and automatic restarts.*
- **Oct 12, validation.** Tested blind on 281 confirmed TESS planets: found 241 (86%). Rejected 70% of proven fakes while keeping 78% of real planets.
- **Oct 12.** Blended neighbour stars and glitches that change from orbit to orbit. *Fix: centroid-shift and dip-consistency tests.*
- **Oct 13.** Six "candidates" were pulsating stars: smooth waves, not dips. *Fix: measure the brightness swing outside the dip.*
- **Oct 13, TIC 415954197.** A software library returned a zero error for the halfway-dip check, hiding a 54-sigma second eclipse. *Fix: Tess measures it directly.*
- **Oct 13, TIC 267889592.** The second eclipse was at phase 0.62, and Tess only looked at 0.5. *Fix: search the whole orbit. 0 of 16 simulated planets wrongly flagged.*
- **Oct 13, deep dive on all 11 candidates.** Every TESS sector, up to six years. 5 binaries, 3 too weak, 3 still open.

Takeaway

Most of building a planet hunter is building an impostor hunter. Out of 2,198 repeating dips, the hard part was the 2,195 that weren't planets.

Reference

NASA's TESS at a glance

Full name	Transiting Exoplanet Survey Satellite
Launched	April 18, 2018, on a SpaceX Falcon 9 from Cape Canaveral, Florida
Led by	MIT, for NASA's Explorer program
Orbit	13.7 days, elongated, in step with the Moon's orbit (2:1)
Cameras	4, each covering 24° × 24°
One "sector"	a 24° × 96° strip of sky, watched for about 27 days
Pixel size on the sky	21 arcseconds
Colours it sees	red to near-infrared, 600 to 1,000 nm
Full-frame images	every 30 min (2018 to 2020), 10 min (2020 to 2022), 200 s (since 2022)
NASA candidates (TOIs)	8,151 as of October 13, 2026

How a transit search works

TESS measures how bright each star is, over and over, for weeks. If a planet's orbit is lined up just right, it passes in front of its star once per orbit and blocks a tiny fraction of the light. Search for a dip that repeats on a perfectly regular schedule, and you've found a candidate.

Then comes the hard part: proving it isn't something else. Eclipsing binaries, blended background stars, pulsating stars and the spacecraft's own quirks all make dips too.

Light-curve vocabulary

- **Light curve:** a star's brightness over time.
- **Transit:** a planet crossing in front of its star.
- **Phase:** where you are in the orbit, from 0 to 1.
- **Folding:** stacking every orbit on top of each other so dips line up.
- **ppm:** parts per million. 10,000 ppm = 1%.
- **TIC number:** a star's ID in the TESS Input Catalog.

TIC 267889592: planet or not?

Dip depth $\delta = 0.0150$ → $R_p = R_\star\sqrt{\delta} = 0.75\ R_\odot \times 0.122 = 0.092\ R_\odot$

$= 0.092 \times 109 \approx$ 10 Earths ($\approx$ 0.9 Jupiter) planet-sized, OK

Orbit: $a^3 = M\,P^2 = 0.87 \times (19.054 / 365.25)^2$ → $a = 0.133$ AU

Expected transit: $T = (P/\pi)\arcsin(R_\star/a) = 6.07\ d \times 0.026 = 0.159\ d \approx$ 3.8 h

measured $\approx$ 4.1 h ✓ fits!

Halfway (phase 0.5): nothing ✓

BUT: second dip at phase 0.615, almost the same depth!!

$e\cos\omega \approx (\pi/2)(0.615 - 0.5) = 1.571 \times 0.115 =$ 0.18 → oval orbit

2 equal dips per orbit → 2 similar stars → ECLIPSING BINARY

Distance: parallax $p = 3.522$ mas → $d = 1/p = 284$ pc $\times$ 3.26 = 926 ly

Period from 12 eclipse times: $P = 19.05420 \pm 0.00005$ d ($\pm$ 4 s!)

Next eclipse: $t_0 + 149\,P = 2461345.83$ BJD → Nov 1, 2026, ~2:58 am ET

Lesson: check every sector before trusting a candidate. And look at the WHOLE orbit.

Next step

Reporting new variable stars

Both binaries appear to be missing from the catalogs astronomers use. The American Association of Variable Star Observers (AAVSO) runs the International Variable Star Index (VSX), where anyone can submit a newly found variable star along with the evidence. These are the details Tess has measured for each one.

TIC 267889592

Position (J2000)	07 41 00.9 −51 10 32
Other ID	Gaia DR3 5493043445356438144
Type	EA (detached eclipsing), eccentric
Period	19.05420 ± 0.00005 d
Epoch of eclipse	BJD 2458506.7565
Depths (TESS band)	~1.8% primary, ~1.6% secondary
Secondary at phase	0.615
Data	TESS sectors 7, 8, 9, 34, 35, 61, 62, 88, 89 (QLP)
Note	Unrelated EB 5.7″ away (P = 0.308 d)

TIC 415954197

Position (J2000)	22 49 27.2 +11 05 48
Other ID	Gaia DR3 2718329812895666816
Type	EA (detached eclipsing)
Period	7.1399 d
Epoch of deep eclipse	BJD 2459831.19 (approx.)
Depths (TESS band)	~3.2% primary, ~1.2% secondary
Secondary at phase	0.50
Data	TESS sectors 56, 83 (QLP)
Note	Brighter star 56″ away; source not yet confirmed

Before submitting

- Search VSX, SIMBAD and the ExoFOP-TESS page for each star once more on the day you submit.
- Ideally get ground-based photometry (a campus or club telescope works) to confirm which star in the pixel is eclipsing.
- Attach the folded light curves from this file and credit the TESS QLP data.